Big Data Analytics in Cloud Computing

Cloud-based data processing architectures, analytics platforms, business value and real-world applications

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What Is Big Data Analytics?

Analyzing datasets too large, fast or varied for traditional tools, using cloud infrastructure that scales on demand. Defined by the 5 Vs:



Volume

Terabytes to petabytes of data



Velocity

Data arrives continuously in real time



Variety

Structured, semi-structured, unstructured



Veracity

Quality and trustworthiness of data



Value

Insight that drives decisions

Why the Cloud for Big Data?



Traditional On-Premises

- Fixed capacity, sized for peak load
- Large upfront hardware spend
- Weeks to provision new clusters
- Storage and compute tightly coupled

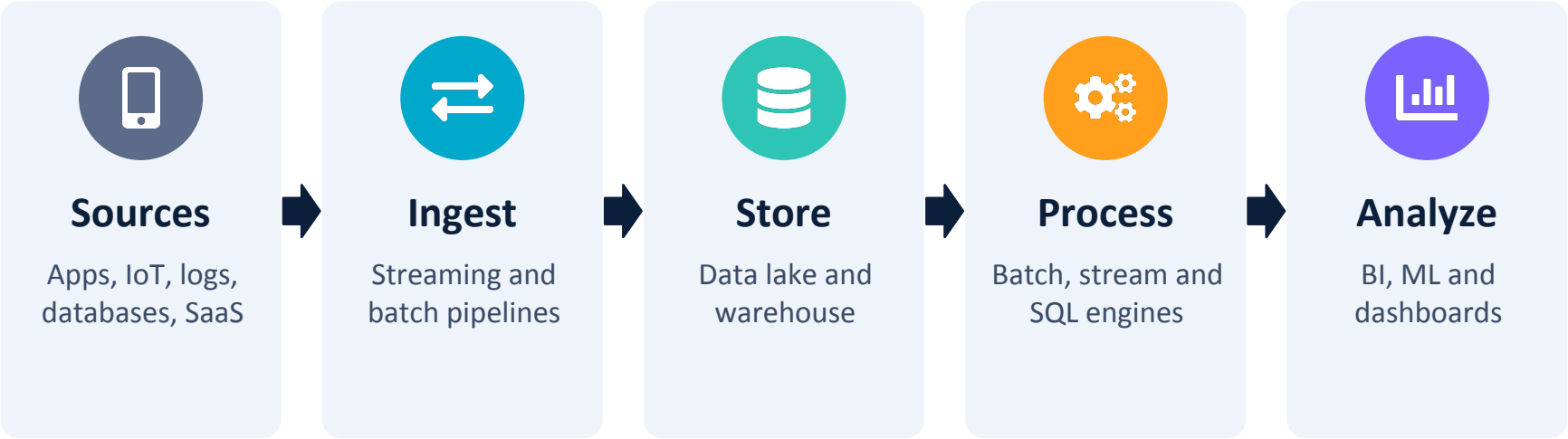


Cloud-Based Analytics

- Elastic: scale up and down on demand
- Pay-as-you-go pricing
- Clusters ready in minutes
- Storage and compute scale independently

Service models: IaaS (raw VMs and storage) • PaaS (managed Spark, warehouses) • SaaS (BI dashboards)

Cloud Data Processing Architecture



Cross-cutting layer: elastic compute • security and access control • governance • orchestration and monitoring

Three Processing Patterns



Batch

Large volumes processed on a schedule

Typical use
Reports, ETL, model training

Tools
Spark, Hadoop, Dataflow



Streaming

Events processed as they arrive

Typical use
Fraud alerts, live monitoring

Tools
Kafka, Flink, Kinesis



Interactive SQL

Fast ad-hoc queries on lake or warehouse

Typical use
Self-service analytics

Tools
BigQuery, Redshift, Synapse

Leading Cloud Analytics Platforms

Capability	AWS	Microsoft Azure	Google Cloud
Data lake storage	Amazon S3	Data Lake Storage	Cloud Storage
Batch processing	EMR, Glue	Databricks, Synapse	Dataproc, Dataflow
Data warehouse	Redshift	Synapse Analytics	BigQuery
Streaming	Kinesis, MSK	Event Hubs	Pub/Sub
ML and BI	SageMaker, QuickSight	Azure ML, Power BI	Vertex AI, Looker

Open-source engines (Spark, Kafka, Flink) run on all three, easing portability.

Business Value



Faster decisions

Real-time dashboards replace week-old reports



Lower cost

Pay for use; no idle hardware or upfront capex



Scale on demand

Handle peak events without re-architecting

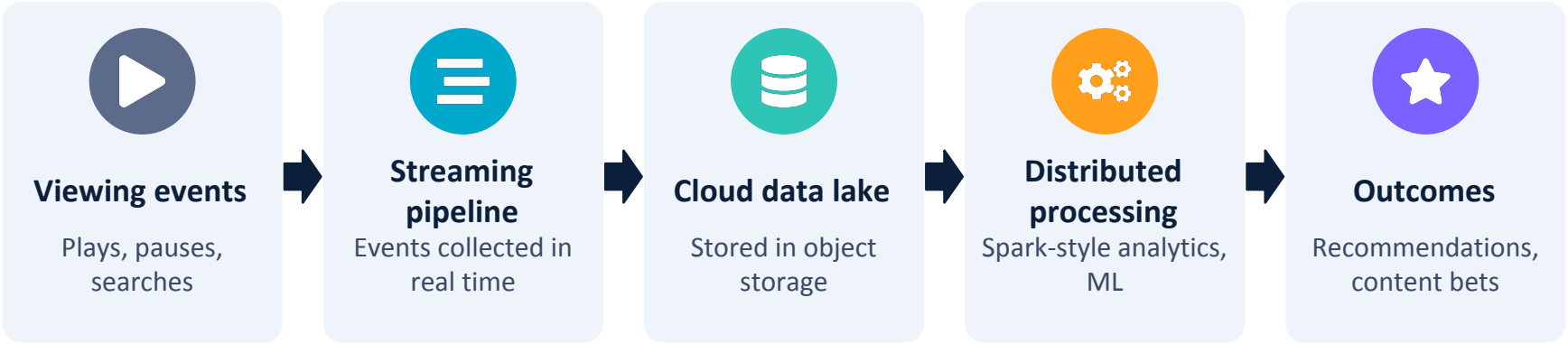


Innovation with AI

Managed ML turns data into predictions

Use Case: Netflix Personalization

Netflix runs on AWS and analyzes viewing behavior at massive scale to recommend titles and guide content decisions. Simplified flow:



Also: Uber (real-time pricing, ETAs) • Banks (fraud detection) • Healthcare (genomics, patient analytics)

Challenges and Best Practices

Security and privacy	Encrypt data, enforce least-privilege access, meet regulations
Cost control	Tag resources, right-size clusters, use storage tiers
Data governance	Catalog data, track lineage, monitor quality
Vendor lock-in	Favor open formats and portable engines

Conclusion and Key Takeaway

Cloud computing makes big data analytics scalable, affordable and fast, turning raw data into a competitive advantage.

Architecture

Ingest, store, process, analyze, all on elastic infrastructure

Platforms

AWS, Azure and Google Cloud offer managed, integrated services

Value

Faster decisions, lower cost and AI-driven innovation

Thank You